**1-to'plam.** *Daraja qoidalari – nol/manfiy ko'rsatkichsiz sof mashq.*

1. $2^2 \times 2^3 =$

2. $\frac{3^4}{3^2} =$

3. $(2^2)^3 =$

4. $5^3 \times 5^2 =$

5. $\frac{4^5}{4^3} =$

6. $(3^2)^2 =$

7. $2^3 \times 2^3 =$

8. $(2^3)^3 =$

**2-to'plam.** *Nol va manfiy ko'rsatkichlar.*

1. $5^0 =$

2. $2^{-3} =$

3. $4^{-2} =$

4. $2^0 \times 2^{-2} =$

5. $3^{-2} \times 3^4 =$

6. $5^2 \times 5^{-2} =$

7. $10^{-1} \times 100 =$

8. $2^{-1} + 2^{-2} =$

**3-to'plam.** *Aralash – haqiqiy test zichligi.*

1. $\dfrac{(2^2 \times 2^3)^2}{2^4} =$

2. $(3 \times 3^2)^2 =$

3. $\dfrac{5^3}{5^2 \times 5^{-1}} =$

4. $(2^{-1})^{-3} =$

5. $(4^2)^0 =$

6. $\dfrac{2^5}{2^{-2}} =$

7. $\dfrac{(3^2)^2}{3^3} =$

8. $10^3 \times 10^{-1} \div 10 =$

**4-to'plam.** *Kvadrat ildizlarni soddalashtirish.*

1. $\sqrt{16} =$

2. $\sqrt{49} =$

3. $\sqrt{18} =$

4. $\sqrt{50} =$

5. $\sqrt{72} =$

6. $\sqrt{20} =$

7. $\sqrt{45} =$

8. $\sqrt{8} =$

**5-to'plam.** *Ildizlarni qo'shish va ayirish.*

1. $\sqrt{8} + \sqrt{2} =$

2. $\sqrt{18} - \sqrt{2} =$

3. $\sqrt{12} + \sqrt{27} =$

4. $\sqrt{50} - \sqrt{8} =$

5. $2\sqrt{3} \times \sqrt{3} =$

6. $\sqrt{2} \times \sqrt{8} =$

7. $3\sqrt{5} + 2\sqrt{5} =$

8. $\sqrt{45} - \sqrt{20} =$

**6-to'plam.** *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $2^3 \times 2^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$2^3 \times 2^3 = 2^9 = 512$."
   Xato va to'g'ri javob: .....

2. $(2^2)^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$(2^2)^3 = 2^{2+3} = 2^5 = 32$."
   Xato va to'g'ri javob: .....

3. $2^{-3}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$2^{-3} = -8$."
   Xato va to'g'ri javob: .....

4. $\sqrt{16+9}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$."
   Xato va to'g'ri javob: .....

5. $\sqrt{32}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$32 = 16 \times 2$ *deb to'g'ri ajratilgan, lekin javobda* $16\sqrt{2}$ *deb yozilgan.*"
   Xato va to'g'ri javob: .....

**7-to'plam.** *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\dfrac{(2^3)^2}{2^4} + \sqrt{16} =$

2. $\sqrt{3^2 \times 2^2} =$

3. $2^{-2} \times 4^2 + \sqrt{9} =$

4. $(\sqrt{2})^4 =$

5. $1 + \sqrt{5^2 - 3^2} =$

**8-to'plam.** *SAT'da qanday uchraydi – daraja va ildiz qoidalarini qo'llash.*

1. $2^3 \times 2^4$ ifodaning qiymati nechaga teng?

   (A) 128 (B) 64

   (C) 16 (D) 81

2. $\dfrac{5^6}{5^2}$ ifodaning qiymati nechaga teng?

   (A) 25 (B) 625

   (C) 125 (D) 15625

3. $(x^3)^4$ ifodasi barcha $x$ uchun qaysi ifodaga teng?

   (A) $x^7$ (B) $x^{64}$

   (C) $x^{12}$ (D) $x^{81}$

4. $7^0 + 3^2$ ifodaning qiymati nechaga teng?

   (A) 9 (B) 16

   (C) 19 (D) 10

5. $4^{-2}$ ifodaning qiymati nechaga teng?

   (A) $\dfrac{1}{16}$ (B) $-8$

   (C) $-16$ (D) 16

6. Agar $a = 2^5$ va $b = 2^3$ bo'lsa, $\dfrac{a}{b}$ nechaga teng?

   (A) 2 (B) 4

   (C) 8 (D) 16

**8-to'plam (davomi).** *SAT'da qanday uchraydi – daraja va ildiz qoidalarini qo'llash.*

7 $\sqrt{98}$ ni eng oddiy ildiz ko'rinishida yozing.

(A) $14\sqrt{2}$ (B) $2\sqrt{49}$

(C) $7\sqrt{2}$ (D) $49\sqrt{2}$

8 Kvadratning yuzi 48 kvadrat birlikka teng. Uning tomoni (eng oddiy ildiz ko'rinishida) nechaga teng?

(A) $16\sqrt{3}$ (B) $6\sqrt{2}$

(C) $2\sqrt{12}$ (D) $4\sqrt{3}$

9 $3^4 \times 3^{-2}$ ifodaning qiymati nechaga teng?

(A) 9 (B) 729

(C) $\frac{1}{9}$ (D) 1

10 $2^n = \frac{1}{8}$ tenglamasini qanoatlantiruvchi $n$ qanday qiymatga ega?

(A) 3 (B) $-3$

(C) $-8$ (D) 8

11 $\sqrt{45} + \sqrt{20}$ ifodaning qiymati nechaga teng?

(A) $5\sqrt{65}$ (B) $\sqrt{65}$

(C) $5\sqrt{5}$ (D) $25\sqrt{5}$

12 Idishdagi bakteriyalar soni har soatda ikki baravar oshadi. Boshlang'ich son $P_0 = 2^3$. Yana 2 marta ikki baravar oshgandan keyin, soni (daraja ko'rinishida) nechaga teng?

(A) $2^6$ (B) $2^1$

(C) $2^9$ (D) $2^5$

## *Qaydlar uchun*

*Nimalar esingizda qoldi?*

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

YouTube

Telegram

# Javoblar

| To'plam | # | Javob | # | Javob |
|---|---|---|---|---|
| 1-to'plam | 1 | 32 | 2 | 9 |
| | 3 | 64 | 4 | 3125 |
| | 5 | 16 | 6 | 81 |
| | 7 | 64 | 8 | 512 |
| 2-to'plam | 1 | 1 | 2 | $\frac{1}{8}$ |
| | 3 | $\frac{1}{16}$ | 4 | $\frac{1}{4}$ |
| | 5 | 9 | 6 | 1 |
| | 7 | 10 | 8 | $\frac{3}{4}$ |
| 3-to'plam | 1 | 64 | 2 | 729 |
| | 3 | 25 | 4 | 8 |
| | 5 | 1 | 6 | 128 |
| | 7 | 3 | 8 | 10 |
| 4-to'plam | 1 | 4 | 2 | 7 |
| | 3 | $3\sqrt{2}$ | 4 | $5\sqrt{2}$ |
| | 5 | $6\sqrt{2}$ | 6 | $2\sqrt{5}$ |
| | 7 | $3\sqrt{5}$ | 8 | $2\sqrt{2}$ |
| 5-to'plam | 1 | $3\sqrt{2}$ | 2 | $2\sqrt{2}$ |
| | 3 | $5\sqrt{3}$ | 4 | $3\sqrt{2}$ |
| | 5 | 6 | 6 | 4 |
| | 7 | $5\sqrt{5}$ | 8 | $\sqrt{5}$ |
| 7-to'plam | 1 | 8 | 2 | 6 |
| | 3 | 7 | 4 | 4 |
| | 5 | 5 | – | – |
| 8-to'plam | 1 | A | 2 | B |
| | 3 | C | 4 | D |
| | 5 | A | 6 | B |
| | 7 | C | 8 | D |
| | 9 | A | 10 | B |
| | 11 | C | 12 | D |
| 6-to'plam | 1. Ko'rsatkichlarni ko'paytirgan, qo'shish kerak edi (ko'paytma qoidasi). To'g'ri: $2^{3+3} = 2^6 = 64$. | | | |
| | 2. Ko'rsatkichlarni qo'shgan, ko'paytirish kerak edi (daraja darajasi). To'g'ri: $2^{2\times3} = 2^6 = 64$. | | | |
| | 3. Manfiy ko'rsatkichni asosni manfiy qiladi deb tushungan. To'g'ri: $2^{-3} = \frac{1}{8}$. | | | |
| | 4. Yig'indining ildizini ildizlar yig'indisiga bo'lgan – bu hech qachon to'g'ri emas. To'g'ri: $\sqrt{25} = 5$. | | | |
| | 5. 16 ni to'g'ri ajratgan, lekin uning ildizini olmagan. To'g'ri: $\sqrt{16 \times 2} = 4\sqrt{2}$. | | | |